UNIVERSITI TEKNOLOGI MALAYSIA

MID TERM TEST

SEMESTER 1, 2022/2023

SUBJECT CODE	**: SECJ 3553 / SCSJ 3553**
SUBJECT	**: ARTIFICIAL INTELLIGENCE**
TIME	**: 2 HOURS 15 MINUTES**
DATE	**: 29 NOV 2022**
PLACE	**: N24 BK1- BK6**

INTRODUCTIONS TO STUDENTS:

This test book consists of 2 sections:

Part A: Multiple Choice Questions [25 Marks]

Part B: Theory and Applications [75 Marks]

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET.

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

SECTION A **TOTAL MARKS: 25**

ARTIFICIAL INTELLIGENCE

Choose the correct answer for the following questions. Each question carries 1 mark.

1. A method that was created to determine whether or not a machine may show artificial intelligence is called as ____________.

    a. Boolean Algebra
    b. Turing Test
    c. Logarithm
    d. Algorithm

2. Which of the following criteria is used to classify artificial intelligence?

    a. Functionally only
    b. Capabilities only
    c. Capabilities and Functionally
    d. It is not categorized

3. Which of the following is an extension of an application for artificial intelligence?

    a. Game Playing
    b. Planning and Scheduling
    c. Diagnosis
    d. All of the above mentioned

4. AI-based financial systems, particularly banking and credit card processing firms can process millions of transactions per minute which enable real-time identification of potentially fraudulent activities whether a given transaction is valid in milliseconds. For example, these systems can determine which purchases don't correspond to typical spending behaviours or examine interactions between payors to determine whether something needs to be flagged for deeper investigation.

    Based on the given statement, which AI domains that best describe the AI-based financial systems?

    a. Heuristic Processing
    b. Cognitive Science
    c. Natural Language Processing
    d. Pattern Recognition

5. Consider the following scenario, which type of artificial intelligent belongs to?

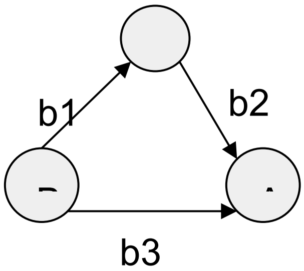
These machines only focus on current scenarios and react on it as per possible best action.

IBM's Deep Blue system is an example of reactive machines.

Such AI systems do not store memories or past experiences for future actions.

a. Limited memory
b. Reactive machines
c. Theory of mind
d. Self-awareness

6.

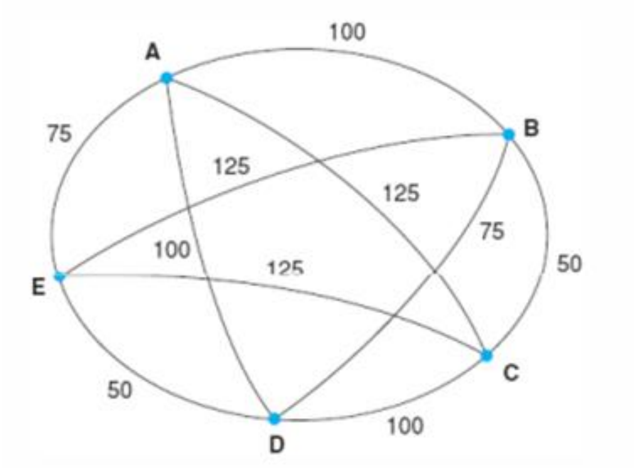


Based on the above diagram, which are the possible arc from node B to node A

i. b1
ii. b1 ⇒ b2
iii. b2
iv. b3

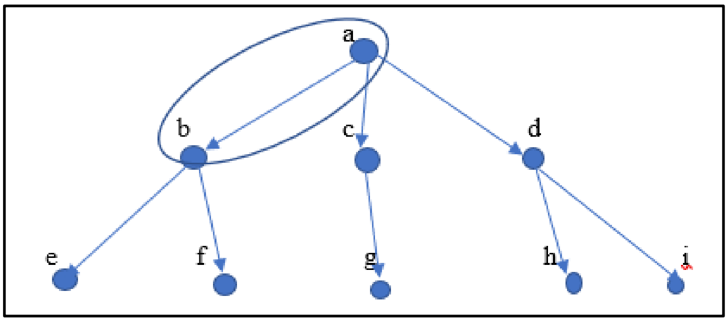
a. i only
b. ii and iii
c. ii and iv
d. ii only

7. Consider the graph below representing the traveling salesperson problem. The node denotes the cities to visit and the edge denotes the distance connected to every city. Which answer is **FALSE** about this graph?



a. The route can be taken from city A, B, C, D to E.
b. When the salesperson starts from city A, the total distance should be counted for every city he visits until he returns to city A.
c. The total number of paths that are possible to be traversed is 4!
d. When the salesperson starts from city C, the next city to be visited can be either B, D, E, or A.

8.



Based on the graph theory above node a (shown in the circle) is the ancestor for which nodes?

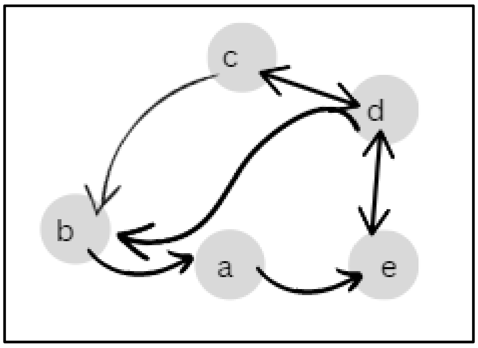
a. Node g
b. Node h
c. Node e and f
d. Node h and i

9. Problem formulation is the process of deciding what actions and states to consider, given a goal. Which of these are not the things that need to be specified during problem formulation?

a. Initial State

b. Actions
c. Goal Test
d. Time complexity

10. Based on the structure of the Graph Theory below, determine the correct arcs based on the nodes a, b, c, d, and e.



a. {(b,a), (b,c), (c,b), (c,d), (d,e), (a,e)}
b. {(b,a), (a,e),(e,d),(d,e),(c,d),(d,c), (c,b), (d,b)}
c. {(b,a), (a,e),(e,d),(d,e),(c,d),(d,c), (c,b), (d,b), (b,d)}
d. {(b,a), (a,e), (e,a),(e,d),(d,e),(c,d),(d,c), (c,b), (d,b)}

11. What is equivalent to $p \rightarrow (q \rightarrow r)$

a. $(p \land q) \rightarrow r$
b. $p \rightarrow (q \lor r)$
c. $\neg q \rightarrow (p \lor r)$
d. $(p \rightarrow r) \land (q \rightarrow r)$

12. Given the truth table below, identify what should be the expression of x

p	¬ q	**x**
T	F	F
T	T	T
F	F	F
F	T	F

a. $\neg (p \rightarrow q)$
b. $p \land \neg q$
c. $p \rightarrow q$
d. $p \leftrightarrow q$

13. Given the knowledge base as below, identify the predicates related to the statement.

jdt(bergson)
hat-trick(bergson,aiman)
score(corbin,safawi)

a. hat-trick and score
b. aiman, corbin, safawi, bergson
c. jdt, hat-trick, score
d. hat-trick, jdt, score, bergson, aiman, safawi, corbin

14. Listed below are the knowledge representation techniques, except for

a. Syntax network
b. Logical representation
c. Frame representation
d. Production rules

15. Based on the properties given below, which of the following is true?

i. $\forall x \forall y \; equal \; to \; \forall y \forall x$
ii. $\exists x \forall y \; equal \; to \; \forall y \exists x$
iii. $\exists x \exists y \; equal \; to \; \exists y \exists x$

a. i only
b. i and ii
c. ii and iii
d. i and iii

16. Let domain of m includes all students, P(m) be the statement "m spends more than 2 hours playing soccer". Express $\forall m \neg$ P(m) quantification in English.

a. There is a student who spends more than 2 hours playing soccer.
b. There is a student who does not spend more than 2 hours playing soccer.
c. All students spend more than 2 hours playing soccer.
d. No student spends more than 2 hours playing soccer.

17. The statement, "At least one of your friends is perfect". Let P(x) be " x is perfect" and let F(x) be "x is your friend" and let the domain be all people.

a. $\forall x \; (F(x) \; \rightarrow \; P(x))$
b. $\forall x \; (F(x) \; \wedge \; P(x))$
c. $\exists x (F(x) \; \wedge \; P(x))$
d. $\exists x (F(x) \; \rightarrow \; P(x))$

18. For propositional logic, which statement is false?

a. The sentences of Propositional logic can have answers other than True or False
b. Each sentence is a declarative sentence.
c. Propositional logic is a knowledge representation technique in AI.
d. Propositional logic is the simplest form of logic where all the statements are made by propositions.

19. Universe of discourse for the variable x is all students.

    C(x): x is taking SECJ 3553

    M(x) : x is a SE major

    Which of the following is the translation of $\forall x\ (M(x)\ \wedge\ C(x))$ in English?

    a. Every student is a SE major and they all are taking SECJ 3553.
    b. There is some SE student who is taking SECJ 3553.
    c. Every student is an SE major or they are taking SECJ 3553.
    d. All students who take SECJ 3553 are not from SE major.

20. Which rule of inference is used in each of these arguments, “If it is Wednesday, then IKEA will be crowded. It is Wednesday. Thus, IKEA is crowded.”

    a. Implementation Elimination
    b. Modus Ponens
    c. And Introduction
    d. Modus Tollens

21. Which search strategy is also called forward chaining approach?

    a. Data-driven search
    b. Uninformed search
    c. Blind search
    d. Goal-directed approach

22.

    There are a large number of rules that match the facts of the problem and thus produce an increasing number of conclusions or goals. Early selection of a goal can eliminates most of these branches, making this search strategy more effective.

    Given a problem with the above requirement, what is the most suitable mentioned search strategy?

    a. Data-driven search
    b. Uninformed search
    c. Blind search
    d. Backward chaining approach

23. Main factors for searching strategy complexity includes the following, **EXCEPT**:

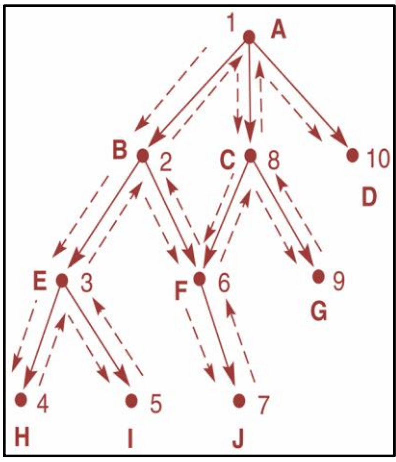
    a. Branching factor
    b. Depth of shallowest goal node
    c. Maximum path length
    d. Optimality of solution found

24. What will be evaluated to calculate the overall cost for the searching strategy?

i. Time complexity
ii. Search cost
iii. Optimal cost
iv. Path cost

a. i and ii
b. iii and iv
c. ii and iv
d. i and iii

25. Given the backtracking graph search and their state space list. The backtracking search starts at iteration 0, root node A. What would be the nodes in State List (SL) in iteration 4?



Initialize: SL = [A]; NSL = [A]; DE = [ ]; CS = A;

AFTER ITERATION	CS	SL	NSL	DE
0	A	[A]	[A]	[ ]
1	B	[B A]	[B C D A]	[ ]
2	E	[E B A]	[E F B C D A]	[ ]
3	H	[H E B A]	[H I E F B C D A]	[ ]
4	I			

a. I E B A
b. H
c. I E F B C D A
d. F B A

SECTION B TOTAL: 75 MARKS

1. Convert the English sentences in parts (I) – (IV) into standard predicate logic sentences when given the following propositions: (8 marks)

A: *teaching AI*

T: *taking* DSA

P: *has facebook*

C: *like cook*

F: *like fishing*

No	English Sentence	Predicate Logic
I	There is a person who is teaching AI and likes to cook.	
II	Every person who is taking DSA has a Facebook page.	
III	It is not the case that everyone taking DSA has a Facebook page or likes to cook.	
IV	Every person who likes fishing will likes to cook.	

2. Convert the following predicate logic formulas into clause form. (5 marks)
   i. (pass(x, exam) ∧ win(x, game)) → happy(x)
   ii. ¬(study(Johan) ∧ pass(Johan))
   iii. (pass(x,exam)) → (holiday(x) ∧ shopping(x))

3. You are given the following statements;

If it is sunny and warm day you will enjoy. If it is warm, you will go cycling. If it is raining there will be no cycling. It is warm day. It is raining. It is sunny.

Using the above statements, construct a proof by ***refutation resolution*** for:

'You will enjoy'.

Use the following notation to represent the problem. (12 marks)

S	**:**	**It is sunny**
W	**:**	**It is warm**
R	**:**	**It is raining**
C	**:**	**Go Cycling**
E	**:**	**Enjoy**

4. Consider the 3-puzzle problem, which is simpler version of the 8-puzzle where the board is 2 X 2 and there are three tiles, numbered 1,2 and 3, and one blank. There are four operators which move the blank up, down, left and right. The start and goal states are given in Figure 1.

Start

2	3
1	

Goal

1	2
	3

Figure 1 : Start and Goal states

i. Construct a state space and show how the path to the goal can be found using breadth-first search. (10 marks)
Note: Use alphabet A, B, C ..to label each node.

ii. Perform a breadth-first search on the state space in c) . List down the order of nodes visited from the starting node to the goal node in a table as shown in Table 1. (10 marks)
* *You may add more row if necessary.*

Table 1

Iteration	**Open**	**Closed**
0	A	-
1		

iii. Briefly explain how the breadth first search (BFS) strategy is implemented. (3 marks)

iv. Breadth first search is a memory-intensive algorithm. Give 2 reasons that contribute to this issue? (2 marks)

5. Figure 2 shows the scenario of the 8-puzzle game using a depth-first search. The start state is denoted by A while the goal state is denoted by N. In depth-first search, the OPEN list comprises a queue state based on the last-in-first-out data structure. Meanwhile, the CLOSED list comprises the states that have been examined and whose children have been generated.

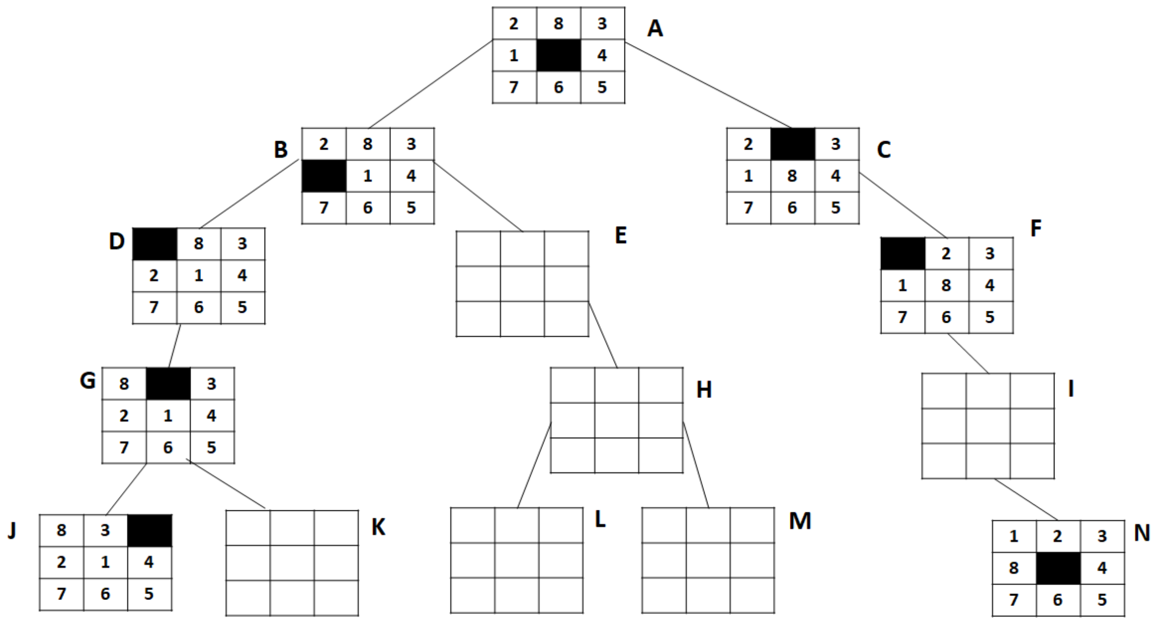


Figure 2: 8-puzzle game using depth first search

Given the information above, answer the following questions:

(a) Draw the remaining states (E, H, I, K, L, and M) in the blank boxes. (6 marks)

(b) Give the sequence of states that will be recorded in the final CLOSED list as follows, CLOSED List = [N, I, ___, ….., B, A] (9 marks)

6. Assume that the location of the water gate is denoted by a letter that has been depicted in Table 2.

Table 2. Node description

Node (Water gate)	**Description**
A	Start node
B, C	The child node of A
B	The parent node of D, E, F
G, H	The child node of C
H	The parent node of I

Based on the given information in Table 2, using depth-first search algorithm, answer the following questions:

(a) Construct a state space graph to represent the above problem. (5 marks)

Answer

(b) Produce the sequence of nodes for the water that can be flowed from gate A to gate G.

(5 marks)